

Reproducing Matcher: Bidirectional Matching as a Robust Outlier Filter

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1. Introduction

Matcher (Liu et al., ICLR 2024) is a training-free framework for one-shot semantic segmentation that composes two frozen vision foundation models — DINOv2 for patch-level features and SAM for class-agnostic segmentation — through three deterministic matching stages [1]. The headline claim is that this composition matches or beats specialist few-shot methods that train directly on the target dataset, without updating any parameter. We reproduce the paper's FSS-1000 result and its critical bidirectional-matching ablation to test the claim, and we report a distributional finding the paper does not surface.

2. Chosen Result

Two numbers from the paper anchor our reproduction. First, the FSS-1000 one-shot result reported in **Table 1**: 87.0 mIoU. Second, the ablation showing the contribution of bidirectional matching reported in **Table 4b**: 81.1 mIoU without reverse matching, an apparent 5.9 mIoU drop. We chose this pair because the bidirectional-matching component is the paper's most-cited methodological contribution, and the ablation directly measures its effect. If we cannot reproduce the gap, the paper's central design choice is unsupported.

3. Methodology

We re-implement the authors' released code with three patches required for compatibility on current Colab Pro: **(i)** `np.int` $\rightarrow$ `int` (NumPy 1.24+ removed the deprecated alias); **(ii)** an environment-variable hook (`DISABLE_REVERSE=1`) that bypasses the cycle-consistency filter, realizing the forward-only ablation by changing one line of behavior with no other hyperparameter touched; and **(iii)** buffered output (`stdbuf -oL python -u`) to handle Drive sync delays during long-running jobs.

All hyperparameters are pinned to the authors' FSS-1000 defaults ($\alpha=0.8$, $\beta=0.2$, $\lambda=1.0$, $K=8$ clusters, top-10 mask merging, coverage threshold 0.3, input 518×518). No tuning. Both regimes use identical settings; only the cycle-consistency filter differs. We evaluate on FSS-1000 fold 0 (20 test classes) with 120 episodes per class — 2,400 total — on a single NVIDIA L4 GPU. Each fold takes 3-7 GPU-hours; a full 12-fold sweep would have been 36-80 hours, well outside what Colab Pro gives reliably. The paper's reported per-fold variance is roughly ± 1 mIoU, so single-fold deltas under 1 mIoU are within their noise band.

4. Results & Analysis

Reproduction is faithful on aggregate metrics, since our results approach the original experiment's. With bidirectional matching: 87.33 mIoU (paper: 87.0). Without: 80.77 mIoU (paper: 81.1). Both within 0.33 mIoU — well inside the paper's per-fold variance. The bidirectional contribution we measure is +6.56 mIoU, slightly larger than the paper's reported +5.9 but in the same direction.

Setting (paper reference)	Ours	Paper	Δ
Matcher full (Table 1, FSS-1000)	87.33	87.0	+0.33
Forward-only ablation (Table 4b)	80.77	81.1	-0.33
Bidirectional contribution	+6.56	+5.9	+0.66

The aggregate hides the actual story. Disaggregated by class, removing reverse matching is not a uniform downward shift. Per-class standard deviation jumps from 8.06 to 19.80, a $2.46\times$ increase. The worst-performing class collapses from 64.9 IoU to 26.1 — a 38.8 absolute IoU loss for that single class. 5 of 20 classes are slightly *better* without bidirectional matching, so the filter is not Pareto-improving either.

This pattern reframes what bidirectional matching does. The paper presents it as a feature improvement, but the per-class evidence says it is a tail-risk reducer: it shields the pipeline from catastrophic failures on a structured subset of visually-ambiguous classes (the ones where DINOv2's patch similarity is biased by background context), at a small cost on classes where forward matching alone would have sufficed. This only means it is only Bias-variance trade-off, not strict improvement.

Reproducibility issue. We initially intended to also reproduce Table 4a (no ILM), but the released code crashes when $\alpha=\beta=0$ — the line `topk_scores / topk_scores.max()` divides by zero, producing NaN downstream. We documented this and pivoted to Table 4b.

5. Reflections

The biggest lesson was that mean mIoU is the wrong summary statistic for an ablation that affects tail behavior. We almost shipped a report that said "we reproduced the +5.9 contribution" — correct but uninformative. The per-class disaggregation took one extra notebook cell and changed the framing of the entire project.

A few engineering takeaways. First, ablation hooks should be one-line gates controlled by environment variables, not forks of the codebase — it makes the main and ablation runs textually identical except for the intervention, which matters for trust. Second, single-fold reproduction is genuinely defensible when the paper reports per-fold variance and the gap to the paper's mean is well inside that variance; we wasted time early on worrying we needed all 12 folds. Third, the released code's Table 4a bug suggests the paper's ablation grid was likely partially cached rather than re-run end-to-end, which is useful context for anyone trying to reproduce work in this area.

What we would do with more time. The 12-fold sweep is the obvious one. Beyond that, we would characterize the failure-mode classes visually — our suspicion is that the catastrophic ones are exactly those where DINOv2's features cluster on background context rather than the object, but we lack qualitative evidence to confirm this. We would also test whether the cycle-consistency idea generalizes to other foundation-model compositions, not just DINOv2 + SAM. The reframing — outlier filtering as the lynchpin — would be more compelling with cross-architecture evidence.

Overall, the headline claim of Matcher reproduces cleanly. The substantive content of our project is the variance story underneath it.

6. References

- [1] Liu, Y., Zhu, M., Li, H., Chen, H., Wang, X., Shen, C. *Matcher: Segment Anything with One-Shot Using All-Purpose Feature Matching*. ICLR 2024. arXiv:2305.13310.
- [2] Oquab, M., Darcet, T., Moutakanni, T., et al. *DINOv2: Learning Robust Visual Features without Supervision*. arXiv:2304.07193, 2023.
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- [4] Li, X., Wei, T., Chen, Y. P., Tai, Y. W., Tang, C. K. *FSS-1000: A 1000-Class Dataset for Few-Shot Segmentation*. CVPR 2020.